

Test-procedure and trial-design choices for the biomarker-treatment interaction in aggregated N-of-1 trials: classical RM-ANOVA, mixed-effects, and the cycle-by-period protocol grid

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1 Abstract

Background. Inference about the biomarker-treatment interaction in aggregated N-of-1 trials depends on two distinct analyst choices: the test procedure (classical repeated-measures ANOVA, linear mixed-effects model, generalised estimating equations) and the trial-design parameters (number of treatment cycles, on-drug and off-drug period lengths, on/off symmetry, optional run-in and run-out phases). Existing methodological treatments hold one of these choices fixed while varying the other. We characterise both axes jointly and quantify the inferential consequences of crossing them.

Methods. We derive the closed-form strict RM-ANOVA F -statistic for the biomarker-stratified treatment-by-time interaction in a balanced split-plot design with categorical predictors and the sphericity assumption, and relate it to the Wald t -test for the corresponding fixed-effects coefficient in a linear mixed model with continuous-time AR(1) residual correlation. We then conduct a Monte Carlo simulation study, designed and reported under the⁷ ADEMP framework, that crosses four test procedures (strict RM-ANOVA, the standard pmsimstats linear-mixed analysis with `corCAR1`, and a generalised estimating equations comparator with naive and with Mancl-DeRouen small-sample sandwich variance) against the biomarker-moderation coefficient $c_{bm} \in \{0, 0.45\}$ and sample size $N \in \{25, 35\}$, a 16-cell factorial at the prazosin-PTSD reference design with $n_{\text{reps}} = 800$ per cell. The cycle-by-period design grid that varies cycle count $k \in \{1, 2, 3, 4, 6\}$, on-drug and off-drug period lengths, and on/off symmetry is specified here but its empirical sweep is deferred to a companion paper.

Results. The closed-form strict RM-ANOVA test agrees with the linear-mixed Wald test under sphericity and balanced design but loses statistical efficiency when within- participant residuals are autocorrelated. In the simulation study

reported here, the linear-mixed analysis dominates strict RM-ANOVA on power for the biomarker-treatment interaction at both sample sizes evaluated, with the power gap widening from approximately 0.19 at $N = 25$ to 0.28 at $N = 35$, while holding nominal or sub-nominal type I error. The generalised estimating equations comparator with the naive sandwich variance inflates type I error to approximately $2.4\times$ nominal at $N = 25$ and $1.5\times$ at $N = 35$; the Mancl-DeRouen small-sample sandwich correction restores nominal control at a moderate power cost. The cycle-by-period design grid that crosses cycle count against period length is specified here and reported in a companion paper.

Conclusions. Trial designers should choose cycle count and period length jointly with the analysis-method choice rather than optimising each separately. The standard pmsimstats linear-mixed analysis is preferred over strict RM-ANOVA across the full parameter grid considered here. We present a test-procedure recommendation at a fixed reference design; the joint cycle-by-period design grid is reported in a companion paper.

2 Introduction

2.1 A claim about how design and test choices interact

[1]

Joint optimisation of test procedure and design parameters is the paper's central argument.

- The two principal choices – test procedure and trial-design parameters – are mutually dependent: validity and power of each test depend on the carryover and autocorrelation structure the design imposes.
- Treating the choices as independent optimisations, as the published literature typically does, produces locally optimal but jointly suboptimal combinations.
- The argument is presented as consequential at the sample sizes characteristic of precision-medicine pilot studies.

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The present paper develops the methodological case for treating the analyst's two principal choices in an aggregated N-of-1 trial jointly rather than separately. The two choices are the test procedure for the biomarker-by-treatment interaction (strict repeated-measures ANOVA, the linear mixed-effects model with continuous-time AR(1) residual correlation, the mixed-effects model for repeated measures, generalised estimating equations) and the trial-design parameters that govern within-subject treatment cycling (number of treatment cycles, on-drug and off-drug period lengths, on/off symmetry, op-

tional run-in and run-out phases). The validity and power of any given test depend on what the design assumes about carryover, autocorrelation, and within-subject variance structure; conversely, the optimal design parameters depend on which test the analyst plans to use. Treating the two as independent optimisations, as the published methodological literature typically does, produces locally optimal but jointly suboptimal combinations, and we shall argue that this gap is consequential at the sample sizes characteristic of precision-medicine pilot studies.

[2]

The paper’s two-part argument establishes a gap in the literature and then fills it at the test-procedure axis.

- No joint design-by-test sensitivity study exists for the aggregated N-of-1 design class.
- The Monte Carlo experiment shows that linear-mixed with AR(1) dominates strict RM-ANOVA wherever residuals depart from compound symmetry.
- Strict RM-ANOVA is retained as a conservative stress-test baseline rather than a primary recommendation.

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The case we develop is two-part. We begin by establishing that a joint design-by-test sensitivity study has not, to the best of our knowledge, been performed for the aggregated N-of-1 design class, and that the absence of one limits the field’s ability to make defensible joint recommendations. We then demonstrate, through a focused Monte Carlo experiment, that the linear mixed-effects test with continuous-time AR(1) residual correlation, used in conjunction with the Hendrickson hybrid open-label-plus-blinded-discontinuation design[?], dominates the strict classical RM-ANOVA test in essentially every parameter cell where within-participant residuals depart from compound symmetry. That is, strict RM-ANOVA remains useful as a stress test — the most stringent baseline that compound-symmetry machinery can deliver — but not as a recommendation in its own right.

2.2 Why the question matters for chronic-condition pharmacology

In this section we shall set out three reasons why the joint treatment of test procedure and design parameters is warranted specifically for the aggregated N-of-1 design class.

[3]